



Sir Syed University
of Engineering & Technology

Database Systems Lab

Project Report

Supervised by
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Database Systems Lab Project

BS Computer Science, Batch 2022F

Employee Management System

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Project Report: Employee Management System

1. Introduction

The Employee Management System (EMS) is a comprehensive solution developed to streamline the management of employee data within an organization. This system is built using C# Windows Forms for the user interface and SQL Server for the database management, ensuring robust and efficient handling of employee-related operations.

2. Project Objective

The primary objective of the EMS is to provide a user-friendly interface for managing employee data, including adding, updating, and deleting employee records, managing projects, and handling salaries. The system aims to enhance productivity by automating routine tasks and providing real-time data access to administrators, managers, and employees.

3. Features

a) User Types

- **Admin:** Has full access to all features, including adding, updating, and deleting employee records, managing salaries, and overseeing projects.
- **Manager:** Can view and manage projects assigned to them and oversee the work of their team members.
- **Employee:** Can view available projects, accept tasks, and update the status of their tickets.

b) Login System

- Employees log in using their employee ID and password.
- Admin ID is fixed at 1000, manager IDs range from 1001 to 1099, and employee IDs start from 1100.
- Only active employees can log in; inactive employees receive an error message.

c) Dashboard

- Displays total employees, active employees, and inactive employees.
- Shows department information and employee data.
- Includes a pie chart representing the total available tickets of current projects.

d) Employee Management

- Admin can add, update, and delete employee records.
- Employee data includes ID, full name, phone number, gender, position, status, password, manager ID, and department ID.

- Positions are predefined and include roles such as Developer, Designer, Consultant, etc.
- Passwords must meet specific security criteria.

e) Salary Management

- Admin can update employee salaries.
- Salary data is linked to the employee record and can be viewed and edited through the salary management interface.

f) Project Management

- **Admin:** Can create, update, and delete projects, assigning them to managers.
- **Manager:** Can view project details, track progress, and oversee team member contributions.
- **Employee:** Can view available projects, accept tasks, and update ticket statuses.

g) Project Details

- Projects have IDs, names, number of tickets, manager IDs, and statuses (available or complete).
- Employees can accept tickets and update their status to complete when finished.
- Managers and employees can track project and ticket statuses through detailed overviews.

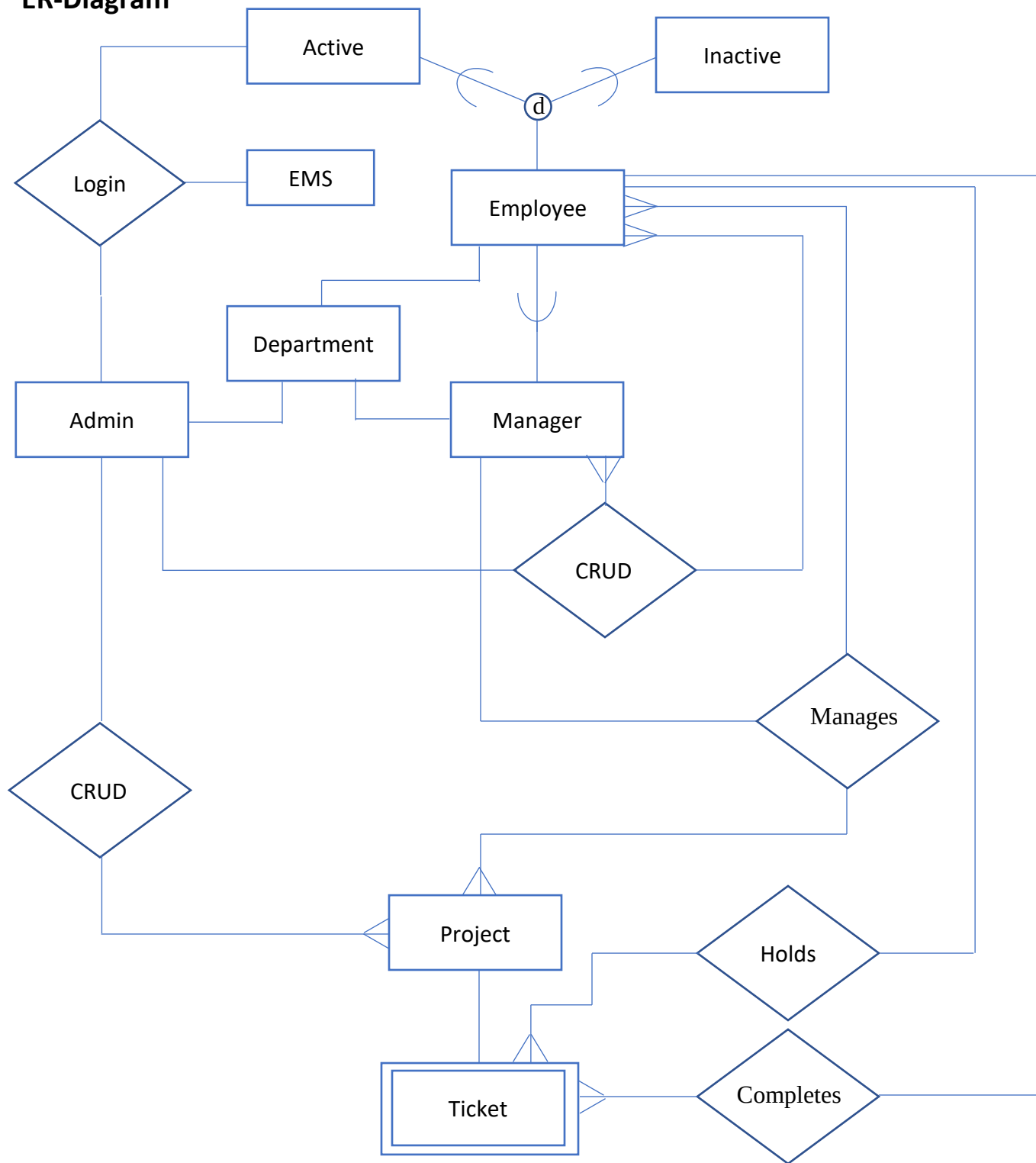
4. Technologies Used

- **C# Windows Forms:** Provides a user-friendly and interactive graphical user interface.
- **SQL Server:** Manages the database, ensuring data integrity and security.
- **Visual Studio:** Integrated development environment used for building the application.

5. Conclusion

The Employee Management System is a robust application designed to facilitate efficient management of employee and project data. By leveraging C# Windows Forms and SQL Server, the system offers a secure and user-friendly interface for administrators, managers, and employees, enhancing overall productivity and organizational management. This project exemplifies the application of modern software development practices in solving real-world business challenges.

ER-Diagram



Schema

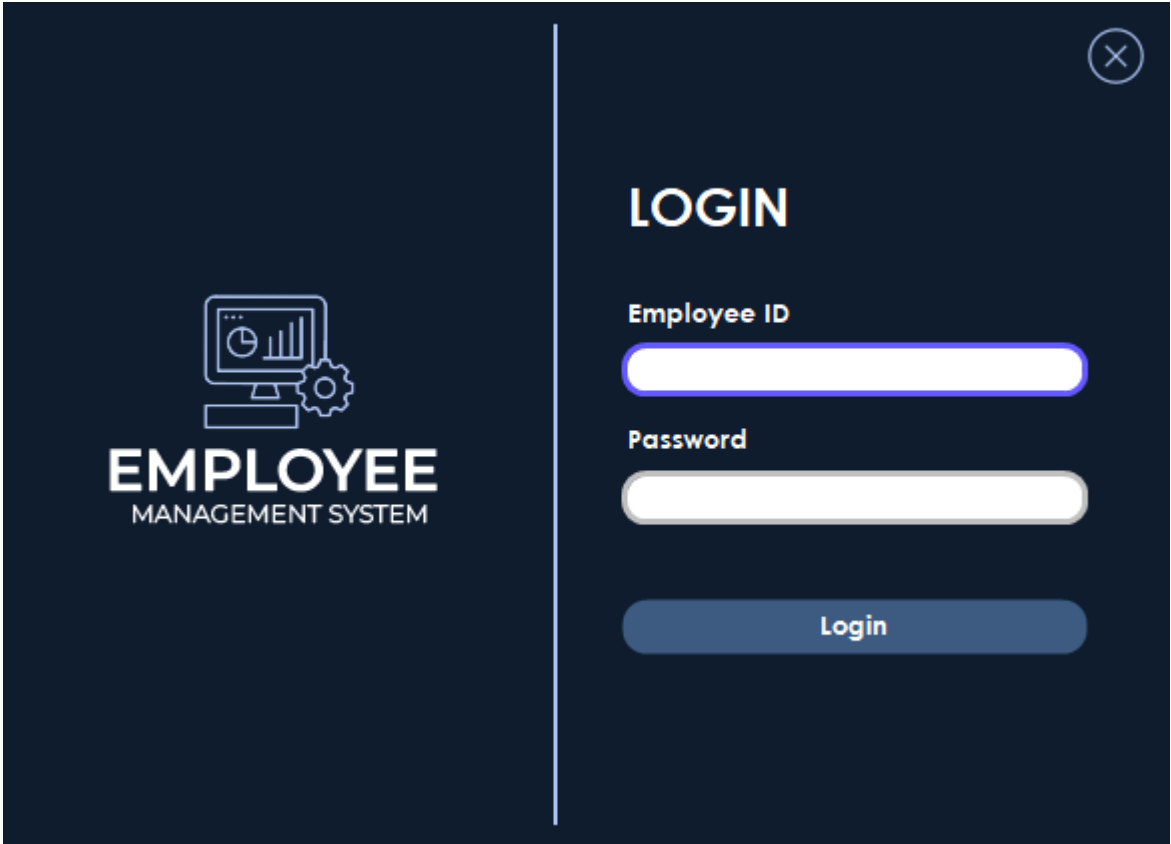
Employee { EMP_ID, emp_name, contact_num, gender, position, salary, insert_date, update_date, delete_date, status, pass, MGR_ID, DEPT_ID (foreign key references Department.DEPT_ID) }

Department{ DEPT_ID, dept_name }

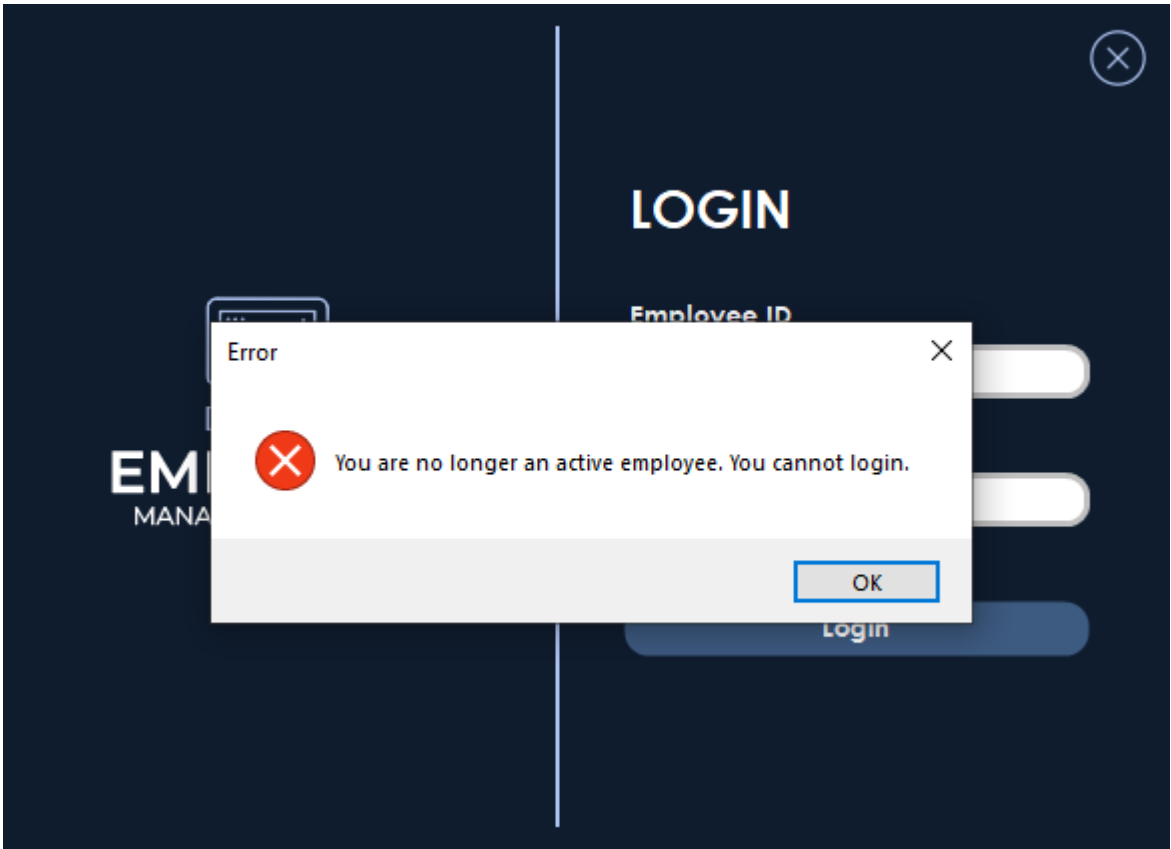
Ticket { TICKET_ID, PROJ_ID (foreign key references Project.PROJ_ID), EMP_ID (foreign key references Employee.EMP_ID), ticket_status }

Project { PROJ_ID, proj_name, total_ticket, completed_ticket, tickets_in_progress, proj_status, EMP_ID (foreign key references Employee.EMP_ID) }

Upon launching the application, the user is presented with a login form. Three types of users can log in: Employee, Admin, and Manager. Only active employees and managers can log in.



The screenshot shows the login interface of the Employee Management System. On the left, there is a logo featuring a computer monitor with a bar chart and a gear, with the text "EMPLOYEE MANAGEMENT SYSTEM" below it. On the right, the word "LOGIN" is displayed in large, bold, white letters. Below this, there are two input fields: "Employee ID" and "Password", both with white borders and rounded corners. A blue "Login" button is positioned below the password field. A small "X" icon in a circle is located in the top right corner of the interface.



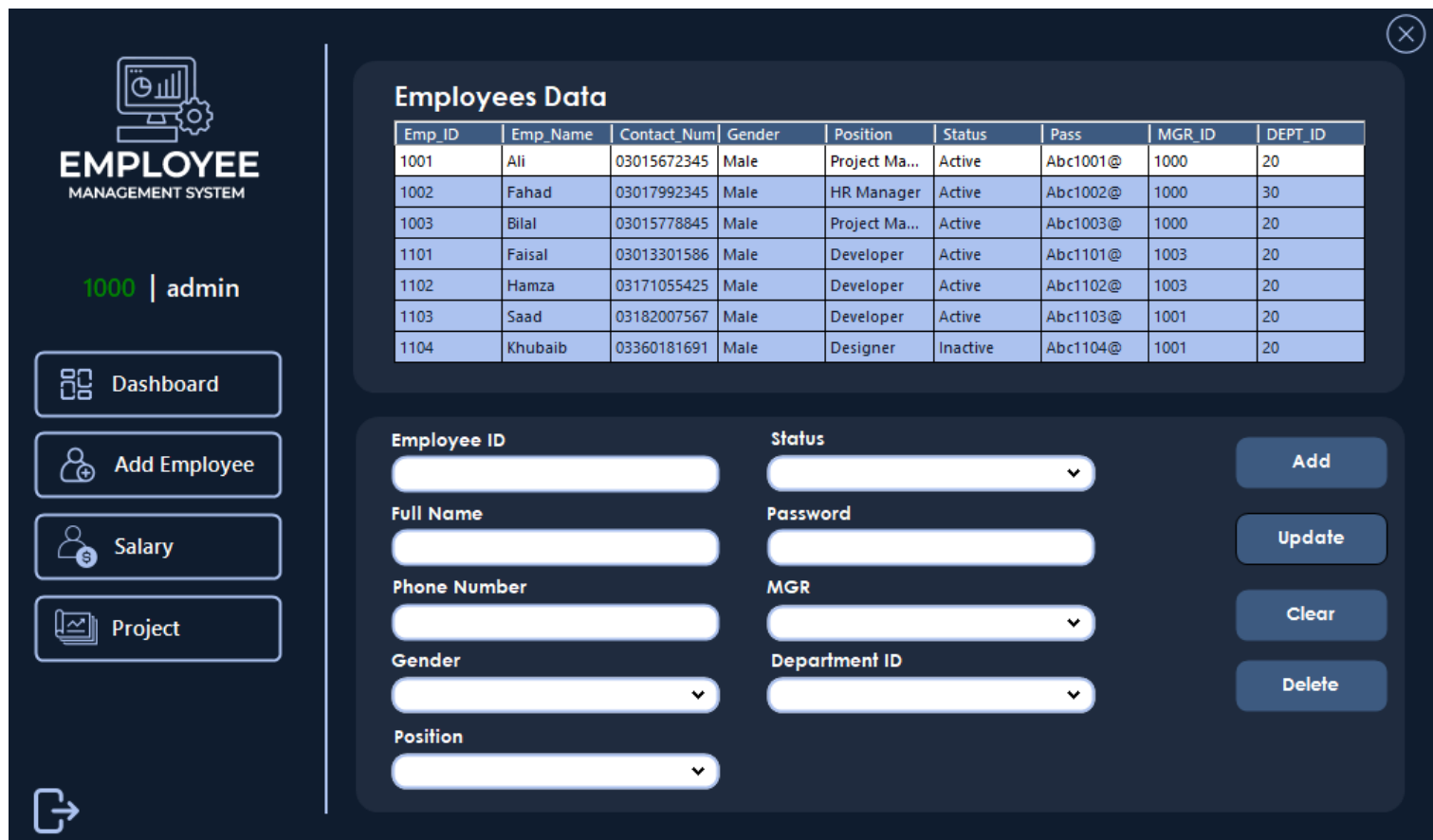
This screenshot shows the same login interface as the previous one, but with an error message dialog box overlaid in the center. The dialog box has a white background and a red "X" icon on the left. The text inside the dialog box reads: "Error" followed by "You are no longer an active employee. You cannot login." and an "OK" button at the bottom right. The background login form is partially obscured by the dialog box.

• Admin Login

Once the admin logs in, the dashboard is displayed. It shows the total number of employees, active employees, and inactive employees. Employee data includes empid, name, position, salary, manager ID, and department ID. Department information is also shown. Additionally, a pie chart displays the total available tickets for projects.



The admin can add, update, delete, and clear employee records. For a manager, the employee ID (empid) must be between 1001 and 1099 inclusive. For an employee, the employee ID (empid) must be 1100 or higher.



The form for adding a new employee includes the following fields and controls:

- Employee ID:** Text input field.
- Status:** Dropdown menu.
- Full Name:** Text input field.
- Password:** Text input field.
- Phone Number:** Text input field.
- MGR:** Dropdown menu.
- Gender:** Dropdown menu.
- Department ID:** Dropdown menu.
- Position:** Dropdown menu.
- Action Buttons:** Add, Update, Clear, Delete.

Employees Data Table:

Emp_ID	Emp_Name	Contact_Num	Gender	Position	Status	Pass	MGR_ID	DEPT_ID
1001	Ali	03015672345	Male	Project Ma...	Active	Abc1001@	1000	20
1002	Fahad	03017992345	Male	HR Manager	Active	Abc1002@	1000	30
1003	Bilal	03015778845	Male	Project Ma...	Active	Abc1003@	1000	20
1101	Faisal	03013301586	Male	Developer	Active	Abc1101@	1003	20
1102	Hamza	03171055425	Male	Developer	Active	Abc1102@	1003	20
1103	Saad	03182007567	Male	Developer	Active	Abc1103@	1001	20
1104	Khubaib	03360181691	Male	Designer	Inactive	Abc1104@	1001	20

Upon clicking the salary button, the salary user control is displayed, allowing the admin to update employee salaries.



EMPLOYEE
MANAGEMENT SYSTEM

1000 | admin

Dashboard

Add Employee

Salary

Project

Employee ID

Full Name

Position

Salary

Update

Clear

Employees Salary

Emp_ID	Emp_Name	Position	Salary
1001	Ali	Project Manager	500000
1002	Fahad	HR Manager	350000
1003	Bilal	Project Manager	550000
1101	Faisal	Developer	450000
1102	Hamza	Developer	450000
1103	Saad	Developer	450000
1104	Khubaib	Designer	400000

Upon clicking the project button, the admin can add, update, and delete projects. Projects can be assigned to managers by selecting a manager ID from a drop-down menu. Once added, project details, including total tickets, completed tickets, tickets in progress, and project status, are displayed in a data grid view.



EMPLOYEE
MANAGEMENT SYSTEM

1000 | admin

Dashboard

Add Employee

Salary

Project

Projects

Proj_ID	Proj_Name	Proj_ticket	completed_ticket	ticket_in_progress	Proj_status	MGR_ID
1	EMS	3	0	0	Available	1001
2	CMS	4	0	0	Available	1003
3	LMS	2	0	0	Available	1003

Create Project

Project ID

Project Name

Number of Tickets

Manager ID

Add

Update

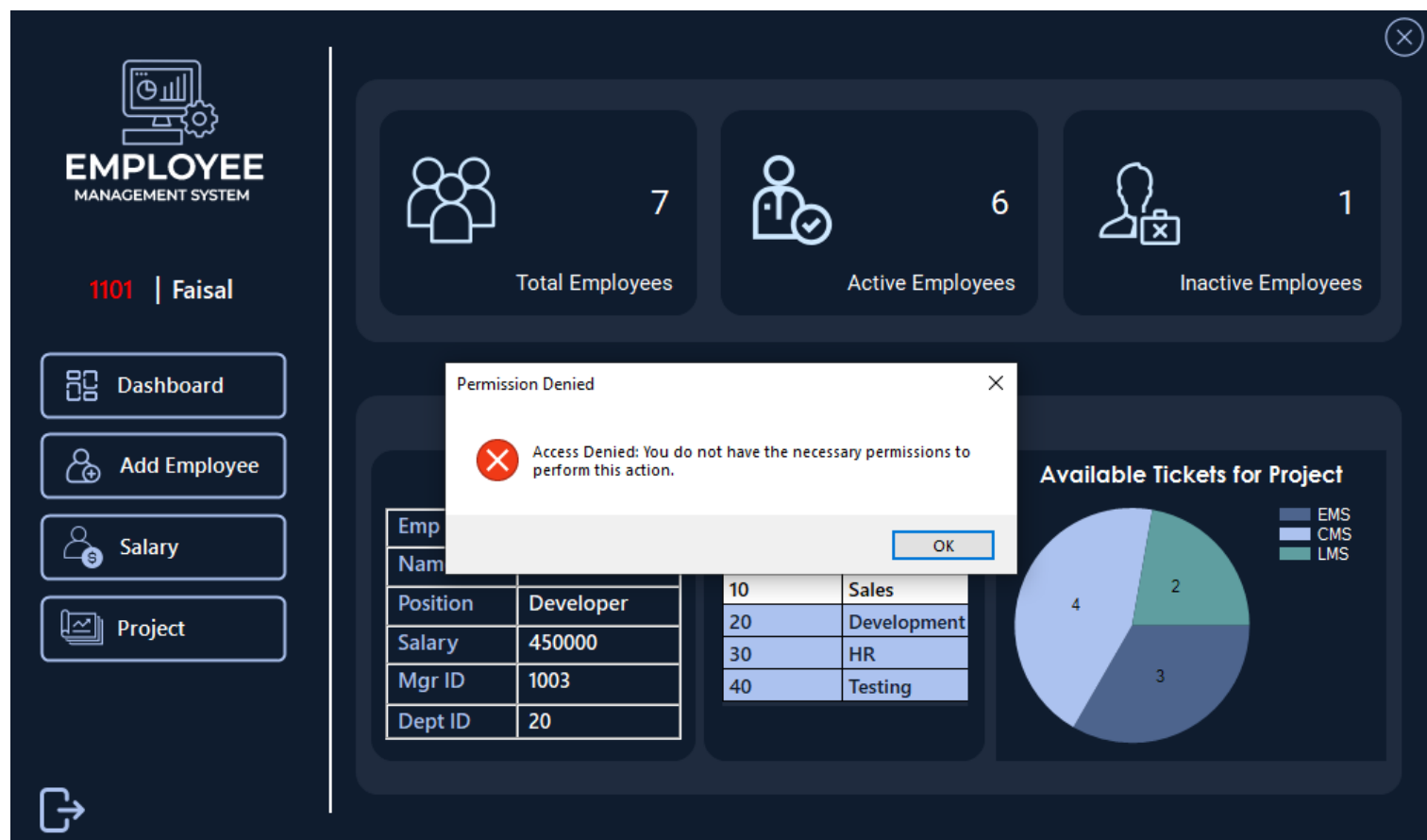
Delete

- **Employee Login**

When an employee logs in, they see a dashboard similar to the admin's dashboard.



Employees do not have permissions to perform CRUD operations on employees. If they click on the "Add Employee" or "Salary" button, an error message is displayed stating "Access denied: You don't have necessary permissions to perform this action."



Employees can choose available project to work on. Clicking a project in the data grid view asks, "Are you sure you want to accept this project?" If confirmed with "Yes," a new ticket is generated with status "Hold," decreasing available tickets by one and increasing tickets in progress by one.



EMPLOYEE MANAGEMENT SYSTEM

1102 | Hamza

Dashboard

Add Employee

Salary

Project

Update Ticket Status

Ticket ID

Completed

Show Cor

Confirmation Message

Are you sure you want to accept this project?

Yes No

Projects

Proj_ID	Proj_Name	Proj_ticket	completed_ticket	ticket_in_progress	Proj_status	MGR_ID
2						
3	LMS	2	0	0	Available	1003

My Projects

ticket_id	Project_id	ticket_status
21	2	Hold

Once the project is completed, the employee can update the ticket status to "Complete." Upon doing so, the count of completed tickets increases by one, and the count of tickets in progress decreases by one.



EMPLOYEE MANAGEMENT SYSTEM

1102 | Hamza

Dashboard

Add Employee

Salary

Project

Update Ticket Status

Ticket ID

Completed

Show Completed Tickets

Projects Available

Proj_ID	Proj_Name	Proj_ticket	completed_ticket	ticket_in_progress	Proj_status	MGR_ID
2	CMS	3	0	1	Available	1003
3	LMS	1	0	1	Available	1003

My Projects

ticket_id	Project_id	ticket_status
21	2	Hold
31	3	Hold

Managers do not have permissions to perform CRUD operations on employees. If they click on the "Add Employee" or "Salary" button, an error message is displayed stating "Access denied: You don't have necessary permissions to perform this action."



EMPLOYEE
MANAGEMENT SYSTEM

1002 | Fahad

Dashboard

Add Employee

Salary

Project



7

Total Employees



6

Active Employees



1

Inactive Employees

Emp

10

20

30

40

1000

30

HR Manager

350000

1000

30

Sales

Development

HR

Testing

Available Tickets for Project



EMS
LMS

Managers manage employees through their project user control, where they can view a project overview. This overview shows which project and ticket ID each employee is working on, along with the ticket's status. Additionally, projects assigned to the manager by the admin are also displayed.



EMPLOYEE
MANAGEMENT SYSTEM

1003 | Bilal

Dashboard

Add Employee

Salary

Project

Projects

Proj_ID	Proj_Name	Proj_ticket	completed_ticket	ticket_in_progres	Proj_status	MGR_ID
2	CMS	1	2	1	Available	1003
3	LMS	0	2	0	complete	1003

Project Overview

TICKET_ID	EMP_ID	EMP_name	Proj_ID	Ticket_status
21	1102	Hamza	2	Hold
22	1101	Faisal	2	complete
23	1101	Faisal	2	complete
31	1102	Hamza	3	complete
32	1101	Faisal	3	Hold